A Cost Estimating Relationship for Satellite Constellations

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Abstract

This paper builds a cost estimating relationship (CER) for satellite constellations. The CER predicts the programme cost from platform mass M and unit count N. The dataset holds 27 systems from 1997 to 2025. Every model input carries a public source. The model writes the programme cost as development cost plus production cost on a learning curve, $\mathrm{TOTAL} = g[\,\mathrm{DEV}(M) + \mathrm{PFM}(M)\, N^{\mathrm{LRE}}]$. Leave-one-out cross-validation gives $R^2 = 0.92$. A second regime (New-Space) costs 5 to 21 times less at equal M and N. Unit-cost data does not identify the one-time development cost. Six programmes with disclosed budgets fix the development-to-production split by class and bound $\mathrm{DEV}(M)$ to a factor of 1.3. On the traditional regime the CER matches the SMAD USCM8+QuickCost benchmark ($R^2 = 0.86$ for both); for New-Space the benchmark over-predicts unit cost by a factor near 8. A planetary factor $F = 1.14$ carries the CER to Mars orbit.

Keywords: cost estimating relationship, satellite constellation, parametric cost model, learning curve

1. Introduction

Public unit-cost data covers few navigation and communication satellites. The cost models in SMAD use data up to the early 2000s [1]. This paper uses 27 systems launched from 1997 to 2025. The CER predicts programme cost from platform mass M and unit count N. Section 2 gives the data and its sources. Section 3 gives the earlier method, the new model, and the planetary factor. Section 4 gives the results and the comparison with the SMAD USCM8 and QuickCost models. Section 5 gives the limits.

2. Data

Table 1 lists the 27 systems and the source of each mass, unit count, and unit cost. AVG is the average unit cost. N is the unit count. M is the platform wet mass. All costs use FY2025 euros. USD converts to EUR at 0.9306 [2]. Earlier-year costs escalate to 2025 with US CPI [3]. For most systems the platform mass is 0.79 times the satellite mass [4]; where the manufacturer publishes the platform mass, the paper uses it directly. The traditional regime holds 22 systems. The New-Space regime holds 5 systems. The paper includes a system when a public per-unit cost exists.

Every cited source was verified twice. A first pass found a public source for each attribute. A second pass assigned one agent per source, which re-opened the source and confirmed that the number appears in it. A source that gives a programme total, from which the paper derives the unit cost as total divided by N, is marked partial († in Table A.3); the derivation is stated. Attributes with no public source are not asserted. New-Space unit costs come from analyst estimates and are labelled. Appendix Appendix A lists every sourced attribute of every system.

Table 1: Dataset. M is the platform wet mass. Reg.: T = traditional, NS = New-Space. † marks a unit cost derived from a sourced programme total; * marks a New-Space analyst estimate.

System	M [kg]	N	AVG [€M]	Reg.
Thales HE-R1000	1000 [5]	5 [6]	106.0 [6]†	T
IRNSS/NavIC	1126 [7]	7 [7]	97.7 [8]†	T
GPS Block III	3065 [9]	10 [9]	183.8 [9]	T
Galileo 2 Generation	1817 [10]	12 [11]	145.0 [11]†	T
GPS Block IIF	1290 [12]	12 [12]	68.2 [13]†	T
GPS Block IIR	1604 [14]	13 [15]	74.4 [16]†	T
Globalstar-3	395 [17]	17 [18]	20.0 [18]	T
O3b MEO	553 [19]	20 [20]	52.0 [21]†	T
Globalstar 2nd Gen	553 [22]	24 [23]	40.6 [24]	T
Galileo FOC Overall	577 [25]	32	44.5 [26]†	T
GLONASS-M	1118 [27]	51 [27]	45.1 [28]†	T
Iridium NEXT	679 [29]	81 [29]	34.2 [30]†	T
Telesat Lightspeed	632 [31]	198 [32]	17.3 [33]†	T
QZSS QZS-2/3/4	3239 [34]	3 [35]	218.2	T
ViaSat-3 F1	5070 [36]	3 [36]	664.4 [37]	T
Inmarsat-6	4321 [38]	2 [39]	368.5 [39]	T
AEHF	7110 [40]	6 [40]	965.0 [40]	T
WGS	4730 [41]	12 [42]	675.6 [41]	T
QZSS QZS-5/6/7	3792 [43]	3 [43]	217.8	T

System	M [kg]	N	AVG [€M]	Reg.
GLONASS-K	739 [44]	11 [45]	110.6 [46]	T
Orbcomm OG2	136	18 [47]	8.9 [48]	T
O3b mPOWER	1343 [49]	13 [50]	158.6	T
OneWeb Gen-1	116 [51]	648 [52]	1.1 [53]*	NS
Amazon Kuiper	434	3232 [54]	1.7 *	NS
Starlink v1.5	242 [55]	4000	0.4 *	NS
Starlink v2-mini	577 [56]	1000	0.8 [57]*	NS
AST BlueBird B2	4819 [58]	48	19.0 [59]*	NS

3. Method

3.1. Earlier method

The earlier method extracted a per-satellite production cost (PFM) from AVG, N, and M. It set the share $\lambda = \text{PFM}/(\text{PFM} + \text{DEV})$ as a linear function of N, from 0.35 at $N = 5$ to 0.15 at $N = 198$. It then fit $\text{PFM} = 0.149\, M^{0.89}$. This method has two limits. First, the extraction removes N; leave-one-out cross-validation of that CER gives $R^2 = 0.62$. Second, the function $\lambda(N)$ comes from a choice, not from data.

3.2. New model

The new model writes the programme cost as

$$\text{TOTAL}(M, N) = F\, g \left[\text{DEV}(M) + \text{PFM}(M)\, N^{\text{LRE}} \right], \tag{1}$$

with $\text{DEV}(M) = d\, M^p$, $\text{PFM}(M) = q\, M^r$, and $\text{LRE} = 1 + \ln(0.9)/\ln 2 = 0.848$. The learning rate is 0.9 [60]. The term g equals 1 for the traditional regime and 10^{off} for New-Space. F is the planetary factor (Section 3.3). The programme cost is development plus production: $\text{TOTAL} = \text{DEV} + \text{PROD}$, with $\text{PROD} = \text{PFM}\, N^{\text{LRE}}$. The average unit cost is $\text{AVG} = \text{TOTAL}/N$, so the identity $\text{TOTAL} = \text{AVG} \cdot N$ holds by construction.

Unit-cost data alone does not identify the one-time development cost DEV. A free fit of d and p gives a development band that runs to zero. The paper fixes this with disclosed budgets. Six programmes report development and production cost separately (Table 2). Their development fraction $\varphi = \text{DEV}/\text{TOTAL}$ is a ratio, so it constrains the split without leaving the dataset's cost basis. The five parameters d, p, q, r, off are fit in one least-squares step against two sets of observations: the unit cost AVG of every system, and the disclosed φ of the five bespoke programmes. This keeps the cross-validated fit and bounds $\text{DEV}(M)$ (Section 4.2). The resolved CER is $\text{DEV}(M) = 3.29\, M^{0.682}$ and $\text{PFM}(M) = 0.215\, M^{0.828}$, with $g = 0.118$ for New-Space.

3.3. Planetary cost factor

The dataset holds Earth-orbit satellites. A Mars-orbit constellation costs more. The paper estimates the increase bottom-up. It scales each bus subsystem by a literature cost-increase factor and by the subsystem's share of bus cost [61, 62]. The sum is a $10.0\% \pm 4.5\%$ increase in bus cost. The paper adopts $F = 1.14$ for Mars and $F = 1$ for Earth. The SMAD QuickCost model applies a single 1.29 planetary factor [63]; the subsystem estimate is lower and traceable.

3.4. Validation

Cross-validation rotates the held-out data. Leave-one-out removes one system, refits, and predicts the removed system. Repeated 5-fold removes a set of systems. Bootstrap resampling gives coefficient intervals. The regime is a design input.

4. Results

4.1. Fit and cross-validation

Leave-one-out gives $R^2 = 0.92$ over the 27 systems and $R^2 = 0.81$ on the traditional regime alone. The 1-sigma prediction band is a factor of 1.7. The typical absolute error is 52%. Figure 1 shows predicted against actual cost.

4.2. Development split

The bootstrap gives DEV(1000 kg) = €365 M with a 1-sigma factor of 1.3. The free fit gave a band from 0 to €510 M. The disclosed fractions bound the split. Development is 57% to 75% of programme cost for the five bespoke programmes and 11% for the one commercial programme. Figure 2 shows the two CERs with their bootstrap bands and the disclosed points.

Table 2: Disclosed development and production cost, FY2025 €M, with source. SAR: Selected Acquisition Report.

Programme	DEV [€M]	Prod [€M]	dev-frac φ	Class	Source
AEHF	9310	7102	0.57	bespoke	[64]
MUOS	5240	3711	0.59	bespoke	[65]
SBIRS	13426	6359	0.68	bespoke	[66]
Galileo	1815	1082	0.63	bespoke	[67]
GPS IIF	2128	719	0.75	bespoke	[68]
WGS	503	4167	0.11	commercial	[69]

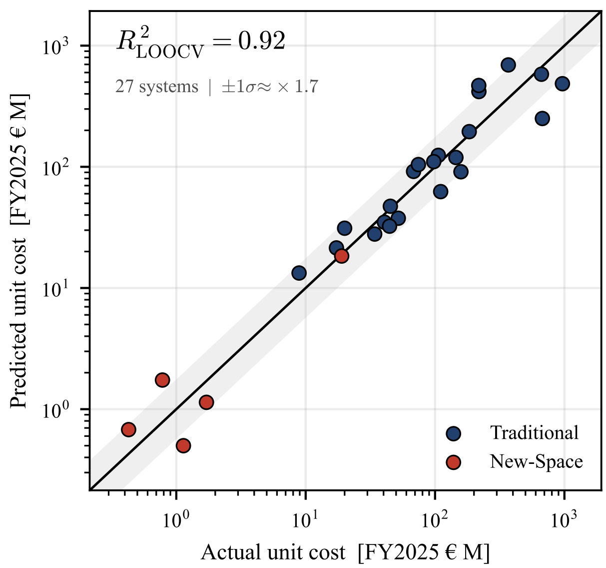

Figure 1: Predicted against actual unit cost. 27 systems. Leave-one-out $R^2 = 0.92$. Band: 1-sigma, factor 1.7.

4.3. New-Space regime

The New-Space regime costs 5 to 21 times less than the traditional regime at equal M and N. The point estimate is 12. The 5 New-Space systems carry masses from 116 to 4819 kg and unit counts from 48 to 4000.

4.4. Comparison with USCM8 and QuickCost

The paper compares the CER with the SMAD USCM8 development and production CERs and the QuickCost payload term [70, 71, 63]. On the traditional regime both models reach the same accuracy: $R^2 = 0.86$, median error 34% for the CER and 29% for USCM8+QuickCost. The USCM8 CERs carry a standard error of 47% for development and 21% for production. On the New-Space regime USCM8+QuickCost over-predicts unit cost by a factor near 8, because it has no term for mass production. Figure 3 shows programme cost over M and N for the three streams. The traditional CER tracks the benchmark; the New-Space regime sits a full order of magnitude below it.

4.5. Operations

The paper carries the SMAD mission operations model for the operations phase [72]. For a 31-satellite constellation over 5 years it gives €159 M.

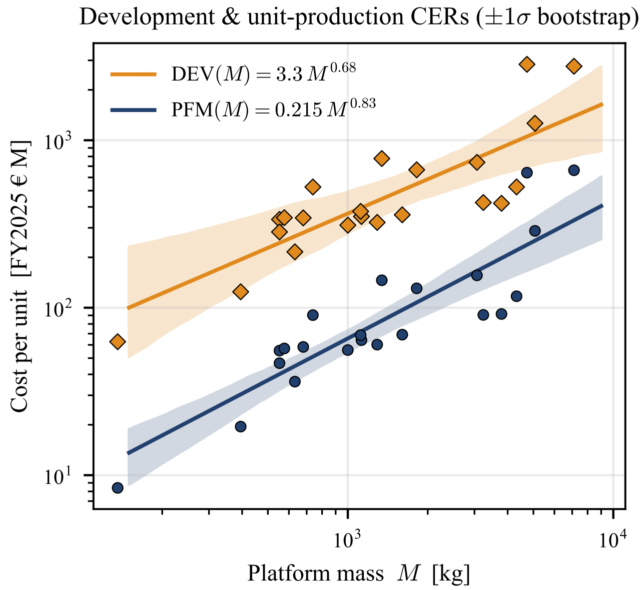

Figure 2: DEV(M) and PFM(M) CER components (lines) with 1-sigma bootstrap bands. Markers: each traditional system's observed programme cost AVG $\cdot N$ split into development (diamonds) and unit production (circles) by the fitted class fraction. The development level is set by the disclosed fractions (Table 2); its band is a factor of 1.3.

The drivers are flight and ground software maintenance, 19 engineers and 4 technicians, and facilities.

5. Limitations

The dataset holds 27 systems. The New-Space regime holds 5 systems; its leave-one-out score is $R^2 = 0.54$. New-Space unit costs come from analyst estimates. The development split depends on programme class. Five of the six anchor programmes are bespoke; the commercial fraction rests on one anchor. Some descriptive attributes (power, spare count) have no public source and are omitted. The escalation to 2025 uses US CPI.

6. Conclusion

The CER predicts constellation programme cost from mass and unit count. Leave-one-out gives $R^2 = 0.92$. A second regime costs 5 to 21 times less. Disclosed budgets fix the development-to-production split and bound

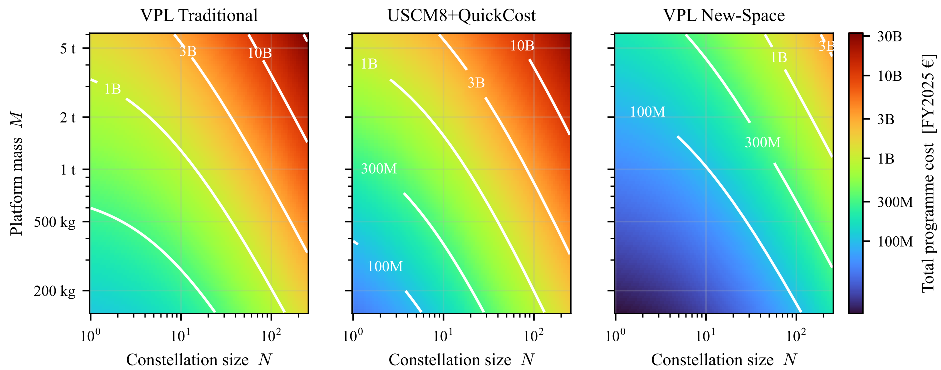

Figure 3: Total programme cost over platform mass and constellation size, with iso-cost contours, for the VPL Traditional CER, the SMAD USCM8+QuickCost benchmark, and the VPL New-Space regime.

the development cost to a factor of 1.3. The CER matches the SMAD USCM8+QuickCost benchmark on traditional systems and corrects its over-prediction on New-Space systems. Every model input is traceable to a public source.

Appendix A. Data provenance

Table A.3 states the source of every sourced attribute of every system. Each value cites the source that contains it. A † marks a partial source: the value is a stated derivation from a sourced total. Attributes with no public source are omitted.

Table A.3: Data provenance: every sourced attribute and its verified source. † marks a partial source (value is a stated derivation, e.g. total/N); * marks a New-Space analyst estimate.

System	Quantity	Value	Source
Thales HE-R1000	Mass	1000 kg (platform)	[5]
	Contract year	2018	[73]
	Units N	5	[6]
	Unit cost	106.0 €M†	[6]†
	Country	EU	[73]
	Purpose	SAR	[73]

System	Quantity	Value	Source
IRNSS/NavIC	Mass	1425 kg	[7]
	Power	1600 W	[7]
	Contract year	2017†	[74]†
	Units N	7	[7]
	Design life	10 yr	[75]
	Spares	2	[76]
	Unit cost	97.7 €M†	[8]†
	Country	India	[7]
	Purpose	Navigation	[76]
GPS Block III	Mass	3880 kg	[9]
	Power	4480 W	[9]
	Contract year	2016	[9]
	Units N	10	[9]
	Design life	15 yr	[15]
	Unit cost	183.8 €M	[9]
	Country	USA	[9]
	Purpose	Navigation	[9]
Galileo 2 Generation	Mass	2300 kg	[10]
	Contract year	2021	[11]
	Units N	12	[11]
	Unit cost	145.0 €M†	[11]†
	Purpose	Navigation	[10]
GPS Block IIF	Mass	1633 kg	[12]
	Power	1952 W	[12]
	Contract year	2016	[12]
	Units N	12	[12]
	Design life	12 yr	[12]
	Unit cost	68.2 €M†	[13]†
	Country	USA	[12]
	Purpose	Navigation	[12]
GPS Block IIR	Mass	2030 kg	[14]
	Units N	13	[15]
	Design life	7.5 yr	[15]
	Unit cost	74.4 €M†	[16]†
Globalstar-3	Mass	500 kg	[17]
	Contract year	2025	[18]
	Units N	17	[18]
	Unit cost	20.0 €M	[18]

System	Quantity	Value	Source
	Purpose	Telecom	[17]
O3b MEO	Mass	700 kg	[19]
	Units N	20	[20]
	Unit cost	52.0 €M†	[21]†
	Country	EU	[77]
	Purpose	Telecom	[77]
Globalstar 2nd Gen	Mass	700 kg	[22]
	Power	2400 W	[22]
	Contract year	2006	[22]
	Units N	24	[23]
	Design life	15 yr	[22]
	Unit cost	40.6 €M	[24]
	Country	USA	[24]
	Purpose	Telecom	[22]
Galileo FOC Overall	Mass	730 kg	[25]
	Power	1600 W	[78]
	Contract year	2010	[25]
	Design life	12 yr	[25]
	Spares	6†	[26]†
	Unit cost	44.5 €M†	[26]†
	Country	EU	[79]
	Purpose	Navigation	[79]
GLONASS-M	Mass	1415 kg	[27]
	Power	1250 W	[27]
	Contract year	2003	[27]
	Units N	51	[27]
	Design life	7 yr	[27]
	Unit cost	45.1 €M†	[28]†
	Country	Russia	[27]
	Purpose	Navigation	[27]
Iridium NEXT	Mass	860 kg	[29]
	Power	2200 W	[29]
	Contract year	2010	[80]
	Units N	81	[29]
	Design life	10 yr	[29]
	Spares	9	[80]
	Unit cost	34.2 €M†	[30]†
	Country	USA	[80]

System	Quantity	Value	Source
	Purpose	Telecom	[29]
Telesat Lightspeed	Mass	800 kg	[31]
	Contract year	2023	[32]
	Units N	198	[32]
	Design life	11 yr	[81]
	Spares	10	[33]
	Unit cost	17.3 €M†	[33]†
	Country	Canada	[32]
	Purpose	Telecom	[33]
QZSS QZS-2/3/4	Sat. mass	4100 kg	[34]
	Units N	3	[35]
	Unit cost	218.2 €M	–
ViaSat-3 F1	Sat. mass	6418 kg	[36]
	Units N	3	[36]
	Unit cost	664.4 €M	[37]
Inmarsat-6	Sat. mass	5470 kg	[38]
	Units N	2	[39]
	Unit cost	368.5 €M	[39]
AEHF	Sat. mass	9000 kg	[40]
	Units N	6	[40]
	Unit cost	965.0 €M	[40]
WGS	Sat. mass	5987 kg	[41]
	Units N	12	[42]
	Unit cost	675.6 €M	[41]
QZSS QZS-5/6/7	Sat. mass	4800 kg	[43]
	Units N	3	[43]
	Unit cost	217.8 €M	–
GLONASS-K	Sat. mass	935 kg	[44]
	Units N	11	[45]
	Unit cost	110.6 €M	[46]
Orbcomm OG2	Sat. mass	172 kg	–
	Units N	18	[47]
	Unit cost	8.9 €M	[48]
O3b mPOWER	Sat. mass	1700 kg	[49]
	Units N	13	[50]
	Unit cost	158.6 €M	–
OneWeb Gen-1	Sat. mass	147 kg	[51]
	Units N	648	[52]

System	Quantity	Value	Source
	Unit cost	1.1 €M*	[53]
Amazon Kuiper	Sat. mass	550 kg	–
	Units N	3232	[54]
	Unit cost	1.7 €M*	–
Starlink v1.5	Sat. mass	306 kg	[55]
	Units N	4000	–
	Unit cost	0.4 €M*	–
Starlink v2-mini	Sat. mass	730 kg	[56]
	Units N	1000	–
	Unit cost	0.8 €M*	[57]
AST BlueBird B2	Sat. mass	6100 kg	[58]
	Units N	48	–
	Unit cost	19.0 €M*	[59]

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